

future assessments. Particular functionality that was missing from the ASAP model includes: ability to model both asymptotic and dome-shaped selectivity; ability to fit length frequency data directly; ability to fit longer time series of data; and, ability to initialize the population assuming a constant historical exploitation rate. The RP encouraged the continued development of ASAP as it provides an accessible software platform that can be used by a wide range of users.

SEDAR 15A Florida Mutton Snapper

Data Workshop Research Recommendations

Life History Workgroup

The biology of *Lutjanus analis* during reproduction remains perhaps the greatest unknown in the life-history of this species. Despite its relatively large body size, exploited status, and gregarious nature during reproduction, the behaviors, location, and sources of individuals of spawning aggregations in Florida and the greater Caribbean remains elusive. Seasonal migration patterns are completely unknown and based on speculation. Primary habitats used by this species during various stages of its ontogeny are undefined. This information would reveal the dependence of the Florida population on various habitats and locations, e.g., a given spawning location; critical information since models have revealed that contributions to the Florida population of *L. analis* in the form of larvae from outside southern sources is minimal (Paris et al. 2005), and that the Florida population is biologically “on its own”. Because of the aforementioned difficulties and differences in staging criteria, we recommend further review of the maturity data from Tequesta and the Florida Keys, and Puerto Rico before accepting the size- and age- at-maturity values from the regressions reported here.

Commercial Workgroup

Increasing the dockside sampling of commercial catches, particularly for the longline and bandit rig fisheries will be important to monitoring the size of fish, areas and depths fished, and fishing effort for this species and other reef fish. The scarcity of otoliths in the earlier portions of the sampling time series restricts the amount of age information that could be used for assessments, and we suggest placing more emphasis on sampling otoliths for this and other reef species to aid future age-structured stock assessments. There is also a need for increasing the amount of discard information (either at-sea or from logbooks) and discard mortality data in modern stock assessments, including this species. Few discards of mutton snapper were actually noted in commercial fishermen’s logbooks, and perhaps the number of fish discarded by commercial fishermen is really low. However, the relatively low frequency of discard logbooks assigned to fishermen may have also been a factor in the low number of discard records provided. Mutton snapper tend to be caught in low numbers with other reef fish species, and relatively few commercial fishing trips actually appear to target this species.

An examination of the conversion factors used to convert landed weight to whole weight should be undertaken. A comparison of the regressions in Life History Section II (Table 2.12) for gutted weight and whole weight would appear to suggest a lower percentage difference between gutted

weight and whole weight at comparable sizes, perhaps as low as 2-5% rather than the 11% currently used for all snappers. However, at this time, there is not enough data to allow a direct comparison of gutted weight to whole weight and derive a suitable conversion factor and the differences suggested would be small and perhaps negligible for the stock assessment.

Ultimately, if allocation between the various sectors of the fishery for mutton snapper and other reef fish are contemplated, conversion factors may become more of an issue.

There were differences noted in the commercial fisheries landings data between the ALS system, the General Canvass data, and the FWC trip ticket data. These differences should be reconciled so that each system will provide comparable numbers where appropriate.

Recreational Workgroup

Biological sampling of recreational landings in Florida has been funded on the West Coast of Florida, including Monroe County, since 2000, but continues to remain unfunded on the East Coast of Florida. Improved biological data collections are essential for making use of the best stock assessment models currently available, and the Recreational Data Working Group recommends funding and implementation of biological data collections in the shore, private boat, and for-hire modes on the east coast of Florida. The Recreational Data Working Group recommends continued funding for discard data collection and improved data collections on depth and area fished in the Headboat At-Sea Survey in Florida. Data on discarded catch is particularly important for size and bag regulate species, such as mutton snapper. The Working Group also recommends better data collection for area and depth fished in the MRFSS. Depth and area fished are particularly important for calculating depth and area-dependent discard mortality rates for reef fish species, such as mutton snapper, that are found in progressively deeper habitats throughout their life history.

Indices Workgroup

GENERAL recommendations: Explore night fish data! No data taken at night by anyone!

Assessment Workshop Research Recommendations

Life History

The maturity analysis used in this assessment was based on only 32 fish. A study should be designed to collect mutton snapper for age and gonad samples at spawning sites during the spawning season. This would entail a multi-year study to identify the diurnal usage patterns at spawning sites during year and to collect gonad samples for histological examination. To maintain quality and ensure consistency among readers, a set of training histological slides should be developed.

Dependent Data Collections

It is essential that adequate numbers of aging structures be collected from all sectors of the fishery from all regions. A weakness of the assessment was the paucity of age samples in the 1980s and early 1990s.

Review Workshop Research Recommendations

The Review Panel replied to Term of Reference 9.

9. Review the research recommendations provided by the Data and Assessment workshops and make any additional recommendations warranted. Clearly indicate the research and monitoring needs that may appreciably improve the reliability of future assessments. Recommend an appropriate interval for the next assessment.

The DW and AW made numerous recommendations regarding further research that might improve future assessments of mutton snapper. The review panel supports those recommendations, and in particular endorses the following:

- Collection of specimens for maturity analysis. After selection criteria had been applied to select an appropriate subset of potential samples, only 32 specimens were available to estimate the maturation schedule for the current assessment,
- Continued monitoring of discards in the commercial and recreational (headboat) fisheries to estimate magnitude and size frequency of discards is endorsed.
- Continuation of the various fishery independent surveys was recommended by the DW. The panel endorses this recommendation, but notes that the current surveys generally encompass only a portion of the habitats and regions of the mutton snapper stock, which may limit their utility for stock assessment. A fishery independent survey that encompasses the range of the stock would have greater value for stock assessment than a multitude of surveys that each are limited in geographic range.

The review panel noted the limited flexibility of the age-structured model (ASAP) used for the mutton snapper assessment and recommends that a more flexible age-based model be used in future assessments. Particular functionality that was missing from the ASAP model includes: ability to model both asymptotic and dome-shaped selectivity; ability to fit length frequency data directly; ability to fit longer time series of data; and, ability to initialize the population assuming a constant historical exploitation rate. The RP encouraged the continued development of ASAP as it provides an accessible software platform that can be used by a wide range of users.